

Blood Pressure in America: A NHANES Analysis, 2013-2018

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2026-09-14

Introduction

The National Health and Nutrition Examination Survey (NHANES) collects medical and demographic information from people in the United States. For this assessment, blood pressure and demographic data from three NHANES survey cycles, 2013–2014, 2015–2016, and 2017–2018, will be prepared for analysis.

Before asking the data to answer any questions, we first need to make sure it is speaking the same language. This report outlines the main data cleaning, manipulation, and tidying steps used to prepare the datasets. The data will be imported, cleaned, reshaped and organized as well as examined for any missing values, unhelpful numbering, and general consistency across the three survey cycles before being combined for further analysis.

Exercise 2: Data import and preliminary exploration

The dimensions of each dataset were examined using the `dim()` function.

```
dim(bp_2013_2014)
```

```
## [1] 9813  23
```

```
dim(bp_2015_2016)
```

```
## [1] 9544  21
```

```
dim(bp_2017_2018)
```

```
## [1] 8704  21
```

```
dim(demo_2013_2014)
```

```
## [1] 10175  47
```

```
dim(demo_2015_2016)
```

```
## [1] 9971 47
```

```
dim(demo_2017_2018)
```

```
## [1] 9254 46
```

The blood pressure datasets contain between 8,704 and 9,813 observations and between 21 and 23 variables. The demographic datasets contain between 9,254 and 10,175 observations and between 46 and 47 variables.

Dataset	Observations	Variables
Blood pressure 2013–2014	9,813	23
Blood pressure 2015–2016	9,544	21
Blood pressure 2017–2018	8,704	21
Demographics 2013–2014	10,175	47
Demographics 2015–2016	9,971	47
Demographics 2017–2018	9,254	46

Variable Types

The structure of each dataset was reviewed using `str()` to identify the types of variables present.

```
str(bp_2013_2014)
```

```
## 'data.frame': 9813 obs. of 23 variables:
## $ SEQN : num 73557 73558 73559 73560 73561 ...
## .. attr(*, "label")= chr "Respondent sequence number"
## $ PEASCST1: num 1 1 1 1 1 1 1 1 1 1 ...
## .. attr(*, "label")= chr "Blood Pressure Status"
## $ PEASCTM1: num 620 766 665 803 949 ...
## .. attr(*, "label")= chr "Blood Pressure Time in Seconds"
## $ PEASCCT1: num NA NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Blood Pressure Comment"
## $ BPXCHR : num NA NA NA NA NA NA 152 NA NA NA ...
## .. attr(*, "label")= chr "60 sec HR (30 sec HR * 2)"
## $ BPAARM : num 1 1 1 1 1 1 NA 1 1 1 ...
## .. attr(*, "label")= chr "Arm selected"
## $ BPACSZ : num 4 4 4 2 3 5 NA 5 4 3 ...
## .. attr(*, "label")= chr "Coded cuff size"
## $ BPXPLS : num 86 74 68 64 92 60 NA 82 86 70 ...
## .. attr(*, "label")= chr "60 sec. pulse (30 sec. pulse * 2)"
## $ BPXPULS : num 1 1 1 1 1 1 1 1 1 1 ...
## .. attr(*, "label")= chr "Pulse regular or irregular?"
## $ BPXPTY : num 1 1 1 1 1 1 NA 1 1 1 ...
## .. attr(*, "label")= chr "Pulse type"
## $ BPXML1 : num 140 170 160 130 170 180 NA 150 140 170 ...
## .. attr(*, "label")= chr "MIL: maximum inflation levels (mm Hg)"
```

```
## $ BPXSY1 : num 122 156 140 108 136 160 NA 118 128 140 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (1st rdg) mm Hg"
## $ BPXDI1 : num 72 62 90 38 86 84 NA 80 74 78 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (1st rdg) mm Hg"
## $ BPAEN1 : num 2 2 2 2 2 NA 2 2 2 ...
## .. attr(*, "label")= chr "Enhancement used first reading"
## $ BPXSY2 : num 114 160 140 102 134 158 NA 124 124 142 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (2nd rdg) mm Hg"
## $ BPXDI2 : num 76 80 76 34 88 82 NA 80 72 78 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (2nd rdg) mm Hg"
## $ BPAEN2 : num 2 2 2 2 1 2 NA 2 2 2 ...
## .. attr(*, "label")= chr "Enhancement used second reading"
## $ BPXSY3 : num 102 156 146 104 142 154 NA 126 114 142 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (3rd rdg) mm Hg"
## $ BPXDI3 : num 74 42 80 38 86 80 NA 82 72 76 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (3rd rdg) mm Hg"
## $ BPAEN3 : num 2 2 2 2 2 NA 2 2 2 ...
## .. attr(*, "label")= chr "Enhancement used third reading"
## $ BPXSY4 : num NA NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Systolic: Blood pres (4th rdg) mm Hg"
## $ BPXDI4 : num NA NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (4th rdg) mm Hg"
## $ BPAEN4 : num NA NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Enhancement used fourth reading"
```

```
str(bp_2015_2016)
```

```
## 'data.frame': 9544 obs. of 21 variables:
## $ SEQN : num 83732 83733 83734 83735 83736 ...
## .. attr(*, "label")= chr "Respondent sequence number"
## $ PEASCCT1: num NA NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Blood Pressure Comment"
## $ BPXCHR : num NA NA NA NA NA NA NA 94 108 NA ...
## .. attr(*, "label")= chr "60 sec HR (30 sec HR * 2)"
## $ BPAARM : num 1 1 1 1 1 1 1 NA NA 1 ...
## .. attr(*, "label")= chr "Arm selected"
## $ BPACSZ : num 4 4 4 5 3 4 2 NA NA 4 ...
## .. attr(*, "label")= chr "Coded cuff size"
## $ BPXPPLS : num 76 72 56 78 76 64 78 NA NA 66 ...
## .. attr(*, "label")= chr "60 sec. pulse (30 sec. pulse * 2)"
## $ BPXPULS : num 1 1 1 1 1 1 1 1 1 1 ...
## .. attr(*, "label")= chr "Pulse regular or irregular?"
## $ BPXPPTY : num 1 1 1 1 1 1 1 NA NA 1 ...
## .. attr(*, "label")= chr "Pulse type"
## $ BPXML1 : num 150 170 160 150 130 140 130 NA NA 140 ...
## .. attr(*, "label")= chr "MIL: maximum inflation levels (mm Hg)"
## $ BPXSY1 : num 128 146 138 132 100 116 102 NA NA 110 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (1st rdg) mm Hg"
## $ BPXDI1 : num 70 88 46 72 70 58 36 NA NA 70 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (1st rdg) mm Hg"
## $ BPAEN1 : num 2 2 2 2 2 2 2 NA NA 2 ...
## .. attr(*, "label")= chr "Enhancement used first reading"
## $ BPXSY2 : num 124 140 132 134 114 122 98 NA NA 112 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (2nd rdg) mm Hg"
```

```
## $ BPXDI2 : num 64 88 44 68 54 58 52 NA NA 74 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (2nd rdg) mm Hg"
## $ BPAEN2 : num 2 2 2 2 2 2 1 NA NA 2 ...
## .. attr(*, "label")= chr "Enhancement used second reading"
## $ BPXSY3 : num 116 134 136 136 98 120 100 NA NA 112 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (3rd rdg) mm Hg"
## $ BPXDI3 : num 62 82 46 70 56 60 56 NA NA 74 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (3rd rdg) mm Hg"
## $ BPAEN3 : num 2 2 2 2 2 2 2 NA NA 2 ...
## .. attr(*, "label")= chr "Enhancement used third reading"
## $ BPXSY4 : num NA NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Systolic: Blood pres (4th rdg) mm Hg"
## $ BPXDI4 : num NA NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (4th rdg) mm Hg"
## $ BPAEN4 : num NA NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Enhancement used fourth reading"
```

```
str(bp_2017_2018)
```

```
## 'data.frame': 8704 obs. of 21 variables:
## $ SEQN : num 93703 93704 93705 93706 93707 ...
## .. attr(*, "label")= chr "Respondent sequence number"
## $ PEASCT1: num NA NA NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Blood Pressure Comment"
## $ BPXCHR : num 120 114 NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "60 sec HR (30 sec HR * 2)"
## $ BPAARM : num NA NA 1 1 1 1 1 NA 1 1 ...
## .. attr(*, "label")= chr "Arm selected"
## $ BPACSZ : num NA NA 4 3 2 3 5 NA 4 3 ...
## .. attr(*, "label")= chr "Coded cuff size"
## $ BPXPLS : num NA NA 52 82 100 68 74 NA 62 68 ...
## .. attr(*, "label")= chr "60 sec. pulse (30 sec. pulse * 2)"
## $ BPXPULS : num 1 1 1 1 1 1 1 1 1 ...
## .. attr(*, "label")= chr "Pulse regular or irregular?"
## $ BPXPITY : num NA NA 1 1 1 1 1 NA 1 1 ...
## .. attr(*, "label")= chr "Pulse type"
## $ BPXML1 : num NA NA 220 140 140 170 140 NA 130 140 ...
## .. attr(*, "label")= chr "MIL: maximum inflation levels (mm Hg)"
## $ BPXSY1 : num NA NA NA 112 128 NA 120 NA 108 112 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (1st rdg) mm Hg"
## $ BPXDI1 : num NA NA NA 74 38 NA 66 NA 68 68 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (1st rdg) mm Hg"
## $ BPAEN1 : num NA NA NA 2 2 NA 2 NA 2 2 ...
## .. attr(*, "label")= chr "Enhancement used first reading"
## $ BPXSY2 : num NA NA NA 114 128 138 118 NA 94 116 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (2nd rdg) mm Hg"
## $ BPXDI2 : num NA NA NA 70 46 78 66 NA 68 72 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (2nd rdg) mm Hg"
## $ BPAEN2 : num NA NA NA 2 2 2 2 NA 2 2 ...
## .. attr(*, "label")= chr "Enhancement used second reading"
## $ BPXSY3 : num NA NA 202 108 128 144 118 NA 102 112 ...
## .. attr(*, "label")= chr "Systolic: Blood pres (3rd rdg) mm Hg"
## $ BPXDI3 : num NA NA 62 76 58 76 68 NA 64 70 ...
## .. attr(*, "label")= chr "Diastolic: Blood pres (3rd rdg) mm Hg"
```

```
## $ BPAEN3 : num NA NA 2 2 2 2 2 NA 2 2 ...
##   .. attr(*, "label")= chr "Enhancement used third reading"
## $ BPXSY4 : num NA NA 198 NA NA 144 NA NA NA NA ...
##   .. attr(*, "label")= chr "Systolic: Blood pres (4th rdg) mm Hg"
## $ BPXDI4 : num NA NA 74 NA NA 74 NA NA NA NA ...
##   .. attr(*, "label")= chr "Diastolic: Blood pres (4th rdg) mm Hg"
## $ BPAEN4 : num NA NA 2 NA NA 2 NA NA NA NA ...
##   .. attr(*, "label")= chr "Enhancement used fourth reading"
```

```
str(demo_2013_2014)
```

```
## 'data.frame': 10175 obs. of 47 variables:
## $ SEQN : num 73557 73558 73559 73560 73561 ...
##   .. attr(*, "label")= chr "Respondent sequence number"
## $ SDDSRVYR: num 8 8 8 8 8 8 8 8 8 ...
##   .. attr(*, "label")= chr "Data release cycle"
## $ RIDSTATR: num 2 2 2 2 2 2 2 2 1 2 ...
##   .. attr(*, "label")= chr "Interview/Examination status"
## $ RIAGENDR: num 1 1 1 1 2 1 1 2 1 2 ...
##   .. attr(*, "label")= chr "Gender"
## $ RIDAGEYR: num 69 54 72 9 73 56 0 61 42 56 ...
##   .. attr(*, "label")= chr "Age in years at screening"
## $ RIDAGEMN: num NA NA NA NA NA NA 5 NA NA NA ...
##   .. attr(*, "label")= chr "Age in months at screening - 0 to 24 mos"
## $ RIDRETH1: num 4 3 3 3 3 1 3 3 2 3 ...
##   .. attr(*, "label")= chr "Race/Hispanic origin"
## $ RIDRETH3: num 4 3 3 3 3 1 3 3 2 3 ...
##   .. attr(*, "label")= chr "Race/Hispanic origin w/ NH Asian"
## $ RIDEXMON: num 1 1 2 1 1 1 2 2 NA 1 ...
##   .. attr(*, "label")= chr "Six month time period"
## $ RIDEXAGM: num NA NA NA NA 119 NA NA 6 NA NA NA ...
##   .. attr(*, "label")= chr "Age in months at exam - 0 to 19 years"
## $ DMQMILIZ: num 1 2 1 NA 2 1 NA 2 2 2 ...
##   .. attr(*, "label")= chr "Served active duty in US Armed Forces"
## $ DMQADFC: num 1 NA 1 NA NA 2 NA NA NA NA ...
##   .. attr(*, "label")= chr "Served in a foreign country"
## $ DMBORN4: num 1 1 1 1 1 1 1 1 1 1 ...
##   .. attr(*, "label")= chr "Country of birth"
## $ DMDCITZN: num 1 1 1 1 1 1 1 1 1 1 ...
##   .. attr(*, "label")= chr "Citizenship status"
## $ DMDYRSUS: num NA NA NA NA NA NA NA NA NA NA ...
##   .. attr(*, "label")= chr "Length of time in US"
## $ DMDEDUC3: num NA NA NA 3 NA NA NA NA NA NA ...
##   .. attr(*, "label")= chr "Education level - Children/Youth 6-19"
## $ DMDEDUC2: num 3 3 4 NA 5 4 NA 5 3 3 ...
##   .. attr(*, "label")= chr "Education level - Adults 20+"
## $ DDMARTL: num 4 1 1 NA 1 3 NA 2 1 3 ...
##   .. attr(*, "label")= chr "Marital status"
## $ RIDEXPRG: num NA NA NA NA NA NA NA NA NA NA ...
##   .. attr(*, "label")= chr "Pregnancy status at exam"
## $ SIALANG : num 1 1 1 1 1 1 1 1 1 1 ...
##   .. attr(*, "label")= chr "Language of SP Interview"
## $ SIAPROXY: num 2 2 2 1 2 2 1 2 2 2 ...
##   .. attr(*, "label")= chr "Proxy used in SP Interview?"
```

```

## $ SIAINTRP: num 2 2 2 2 2 2 2 2 2 ...
##   ..- attr(*, "label")= chr "Interpreter used in SP Interview?"
## $ FIALANG : num 1 1 1 1 1 1 1 1 1 ...
##   ..- attr(*, "label")= chr "Language of Family Interview"
## $ FIAPROXY: num 2 2 2 2 2 2 2 2 2 ...
##   ..- attr(*, "label")= chr "Proxy used in Family Interview?"
## $ FIAINTRP: num 2 2 2 2 2 2 2 2 2 ...
##   ..- attr(*, "label")= chr "Interpreter used in Family Interview?"
## $ MIALANG : num 1 1 1 1 1 1 NA 1 NA 1 ...
##   ..- attr(*, "label")= chr "Language of MEC Interview"
## $ MIAPROXY: num 2 2 2 2 2 2 NA 2 NA 2 ...
##   ..- attr(*, "label")= chr "Proxy used in MEC Interview?"
## $ MIAINTRP: num 2 2 2 2 2 2 NA 2 NA 2 ...
##   ..- attr(*, "label")= chr "Interpreter used in MEC Interview?"
## $ AIALANGA: num 1 1 NA 1 NA 1 NA 1 NA 1 ...
##   ..- attr(*, "label")= chr "Language of ACASI Interview"
## $ DMDHHSIZ: num 3 4 2 4 2 1 3 1 4 7 ...
##   ..- attr(*, "label")= chr "Total number of people in the Household"
## $ DMDFMSIZ: num 3 4 2 4 2 1 3 1 4 7 ...
##   ..- attr(*, "label")= chr "Total number of people in the Family"
## $ DMDHHSZA: num 0 0 0 0 0 0 1 0 1 3 ...
##   ..- attr(*, "label")= chr "# of children 5 years or younger in HH"
## $ DMDHHSZB: num 0 2 0 2 0 0 0 0 1 0 ...
##   ..- attr(*, "label")= chr "# of children 6-17 years old in HH"
## $ DMDHHSZE: num 2 0 2 0 2 0 0 1 0 0 ...
##   ..- attr(*, "label")= chr "# of adults 60 years or older in HH"
## $ DMDHRGND: num 1 1 1 1 1 1 1 2 1 2 ...
##   ..- attr(*, "label")= chr "HH ref person's gender"
## $ DMDHRAGE: num 69 54 72 33 78 56 31 61 42 56 ...
##   ..- attr(*, "label")= chr "HH ref person's age in years"
## $ DMDHRBR4: num 1 1 1 1 1 1 1 1 1 1 ...
##   ..- attr(*, "label")= chr "HH ref person's country of birth"
## $ DMDHREDU: num 3 3 4 3 5 4 5 5 3 3 ...
##   ..- attr(*, "label")= chr "HH ref person's education level"
## $ DMDHRMAR: num 4 1 1 1 1 3 1 2 1 3 ...
##   ..- attr(*, "label")= chr "HH ref person's marital status"
## $ DMDHSEDU: num NA 1 3 4 5 NA 5 NA 5 NA ...
##   ..- attr(*, "label")= chr "HH ref person's spouse's education level"
## $ WTINT2YR: num 13281 23682 57215 55201 63710 ...
##   ..- attr(*, "label")= chr "Full sample 2 year interview weight"
## $ WTMEC2YR: num 13481 24472 57193 55767 65542 ...
##   ..- attr(*, "label")= chr "Full sample 2 year MEC exam weight"
## $ SDMVPSU : num 1 1 1 2 2 1 1 1 2 1 ...
##   ..- attr(*, "label")= chr "Masked variance pseudo-PSU"
## $ SDMVSTRA: num 112 108 109 109 116 111 105 114 106 112 ...
##   ..- attr(*, "label")= chr "Masked variance pseudo-stratum"
## $ INDHHIN2: num 4 7 10 9 15 9 15 10 15 4 ...
##   ..- attr(*, "label")= chr "Annual household income"
## $ INDFMIN2: num 4 7 10 9 15 9 15 10 15 4 ...
##   ..- attr(*, "label")= chr "Annual family income"
## $ INDFMPIR: num 0.84 1.78 4.51 2.52 5 4.79 5 5 0.48 ...
##   ..- attr(*, "label")= chr "Ratio of family income to poverty"

```

```
str(demo_2015_2016)
```

```
## 'data.frame': 9971 obs. of 47 variables:
## $ SEQN : num 83732 83733 83734 83735 83736 ...
## .. attr(*, "label")= chr "Respondent sequence number"
## $ SDDSRVYR: num 9 9 9 9 9 9 9 9 9 ...
## .. attr(*, "label")= chr "Data release cycle"
## $ RIDSTATR: num 2 2 2 2 2 2 2 2 2 ...
## .. attr(*, "label")= chr "Interview/Examination status"
## $ RIAGENDR: num 1 1 1 2 2 2 2 1 1 1 ...
## .. attr(*, "label")= chr "Gender"
## $ RIDAGEYR: num 62 53 78 56 42 72 11 4 1 22 ...
## .. attr(*, "label")= chr "Age in years at screening"
## $ RIDAGEMN: num NA NA NA NA NA NA NA NA 13 NA ...
## .. attr(*, "label")= chr "Age in months at screening - 0 to 24 mos"
## $ RIDRETH1: num 3 3 3 3 4 1 1 3 2 4 ...
## .. attr(*, "label")= chr "Race/Hispanic origin"
## $ RIDRETH3: num 3 3 3 3 4 1 1 3 2 4 ...
## .. attr(*, "label")= chr "Race/Hispanic origin w/ NH Asian"
## $ RIDEXMON: num 1 1 2 2 2 1 1 2 2 1 ...
## .. attr(*, "label")= chr "Six month time period"
## $ RIDEXAGM: num NA NA NA NA NA NA NA 141 54 14 NA ...
## .. attr(*, "label")= chr "Age in months at exam - 0 to 19 years"
## $ DMQMILIZ: num 2 2 1 2 2 2 NA NA NA 2 ...
## .. attr(*, "label")= chr "Served active duty in US Armed Forces"
## $ DMQADFC : num NA NA 2 NA NA NA NA NA NA NA ...
## .. attr(*, "label")= chr "Served in a foreign country"
## $ DMBORN4: num 1 2 1 1 1 2 1 1 1 1 ...
## .. attr(*, "label")= chr "Country of birth"
## $ DMDCITZN: num 1 2 1 1 1 2 1 1 1 1 ...
## .. attr(*, "label")= chr "Citizenship status"
## $ DMDYRSUS: num NA 7 NA NA NA 2 NA NA NA NA ...
## .. attr(*, "label")= chr "Length of time in US"
## $ DMDEDUC3: num NA NA NA NA NA NA NA 6 NA NA NA ...
## .. attr(*, "label")= chr "Education level - Children/Youth 6-19"
## $ DMDEDUC2: num 5 3 3 5 4 2 NA NA NA 4 ...
## .. attr(*, "label")= chr "Education level - Adults 20+"
## $ DMDMARTL: num 1 3 1 6 3 4 NA NA NA 5 ...
## .. attr(*, "label")= chr "Marital status"
## $ RIDEXPRG: num NA NA NA NA 1 NA NA NA NA NA ...
## .. attr(*, "label")= chr "Pregnancy status at exam"
## $ SIALANG : num 1 1 1 1 1 2 1 1 1 1 ...
## .. attr(*, "label")= chr "Language of SP Interview"
## $ SIAPROXY: num 2 2 2 2 2 2 1 1 1 2 ...
## .. attr(*, "label")= chr "Proxy used in SP Interview?"
## $ SIAINTRP: num 2 2 2 2 2 1 2 2 2 2 ...
## .. attr(*, "label")= chr "Interpreter used in SP Interview?"
## $ FIALANG : num 1 1 1 1 1 1 1 1 2 1 ...
## .. attr(*, "label")= chr "Language of Family Interview"
## $ FIAPROXY: num 2 2 2 2 2 2 2 2 2 2 ...
## .. attr(*, "label")= chr "Proxy used in Family Interview?"
## $ FIAINTRP: num 2 2 2 2 2 2 2 2 2 2 ...
## .. attr(*, "label")= chr "Interpreter used in Family Interview?"
```

```

## $ MIALANG : num 1 1 1 1 1 1 NA NA 1 ...
##   .. attr(*, "label")= chr "Language of MEC Interview"
## $ MIAPROXY: num 2 2 2 2 2 2 NA NA 2 ...
##   .. attr(*, "label")= chr "Proxy used in MEC Interview?"
## $ MIAINTRP: num 2 2 2 2 2 2 NA NA 2 ...
##   .. attr(*, "label")= chr "Interpreter used in MEC Interview?"
## $ AIALANGA: num 1 1 NA 1 1 NA 1 NA NA 1 ...
##   .. attr(*, "label")= chr "Language of ACASI Interview"
## $ DMDHHSIZ: num 2 1 2 1 5 5 5 5 7 3 ...
##   .. attr(*, "label")= chr "Total number of people in the Household"
## $ DMDFMSIZ: num 2 1 2 1 5 5 5 5 7 3 ...
##   .. attr(*, "label")= chr "Total number of people in the Family"
## $ DMDHHSZA: num 0 0 0 0 0 1 0 2 2 0 ...
##   .. attr(*, "label")= chr "# of children 5 years or younger in HH"
## $ DMDHHSZB: num 0 0 0 0 2 1 2 1 2 0 ...
##   .. attr(*, "label")= chr "# of children 6-17 years old in HH"
## $ DMDHHSZE: num 1 0 2 0 0 1 1 0 0 0 ...
##   .. attr(*, "label")= chr "# of adults 60 years or older in HH"
## $ DMDHRGND: num 1 1 2 2 2 1 2 2 2 2 ...
##   .. attr(*, "label")= chr "HH ref person's gender"
## $ DMDHRAGE: num 62 53 79 56 42 34 68 35 48 47 ...
##   .. attr(*, "label")= chr "HH ref person's age in years"
## $ DMDHRBR4: num 1 2 1 1 1 2 2 1 2 1 ...
##   .. attr(*, "label")= chr "HH ref person's country of birth"
## $ DMDHREDU: num 5 3 3 5 4 5 1 5 9 4 ...
##   .. attr(*, "label")= chr "HH ref person's education level"
## $ DMDHRMAR: num 1 3 1 6 3 1 2 1 5 1 ...
##   .. attr(*, "label")= chr "HH ref person's marital status"
## $ DMDHSEDU: num 3 NA 3 NA NA 5 NA 5 NA 5 ...
##   .. attr(*, "label")= chr "HH ref person's spouse's education level"
## $ WTINT2YR: num 134671 24329 12400 102718 17628 ...
##   .. attr(*, "label")= chr "Full sample 2 year interview weight"
## $ WTMEC2YR: num 135630 25282 12576 102079 18235 ...
##   .. attr(*, "label")= chr "Full sample 2 year MEC exam weight"
## $ SDMVPSU : num 1 1 1 1 2 1 1 2 1 2 ...
##   .. attr(*, "label")= chr "Masked variance pseudo-PSU"
## $ SDMVSTRA: num 125 125 131 131 126 128 120 124 119 128 ...
##   .. attr(*, "label")= chr "Masked variance pseudo-stratum"
## $ INDHHIN2: num 10 4 5 10 7 14 6 15 77 7 ...
##   .. attr(*, "label")= chr "Annual household income"
## $ INDFMIN2: num 10 4 5 10 7 14 6 15 77 7 ...
##   .. attr(*, "label")= chr "Annual family income"
## $ INDFMPIR: num 4.39 1.32 1.51 5 1.23 2.82 1.18 4.22 NA 2.08 ...
##   .. attr(*, "label")= chr "Ratio of family income to poverty"

```

```
str(demo_2017_2018)
```

```

## 'data.frame': 9254 obs. of 46 variables:
## $ SEQN : num 93703 93704 93705 93706 93707 ...
##   .. attr(*, "label")= chr "Respondent sequence number"
## $ SDDSRVYR: num 10 10 10 10 10 10 10 10 10 10 ...
##   .. attr(*, "label")= chr "Data release cycle"
## $ RIDSTATR: num 2 2 2 2 2 2 2 2 2 2 ...
##   .. attr(*, "label")= chr "Interview/Examination status"

```

```

## $ RIAGENDR: num 2 1 2 1 1 2 2 2 1 1 ...
##   ..- attr(*, "label")= chr "Gender"
## $ RIDAGEYR: num 2 2 66 18 13 66 75 0 56 18 ...
##   ..- attr(*, "label")= chr "Age in years at screening"
## $ RIDAGEMN: num NA NA NA NA NA NA NA 11 NA NA ...
##   ..- attr(*, "label")= chr "Age in months at screening - 0 to 24 mos"
## $ RIDRETH1: num 5 3 4 5 5 5 4 3 5 1 ...
##   ..- attr(*, "label")= chr "Race/Hispanic origin"
## $ RIDRETH3: num 6 3 4 6 7 6 4 3 6 1 ...
##   ..- attr(*, "label")= chr "Race/Hispanic origin w/ NH Asian"
## $ RIDEXMON: num 2 1 2 2 2 2 1 2 2 2 ...
##   ..- attr(*, "label")= chr "Six month time period"
## $ RIDEXAGM: num 27 33 NA 222 158 NA NA 13 NA 227 ...
##   ..- attr(*, "label")= chr "Age in months at exam - 0 to 19 years"
## $ DMQMILIZ: num NA NA 2 2 NA 2 2 NA 2 2 ...
##   ..- attr(*, "label")= chr "Served active duty in US Armed Forces"
## $ DMQADFC : num NA NA NA NA NA NA NA NA NA NA ...
##   ..- attr(*, "label")= chr "Served in a foreign country"
## $ DMBORN4: num 1 1 1 1 1 2 1 1 2 2 ...
##   ..- attr(*, "label")= chr "Country of birth"
## $ DMCITZN: num 1 1 1 1 1 1 1 1 1 2 ...
##   ..- attr(*, "label")= chr "Citizenship status"
## $ DMDYRSUS: num NA NA NA NA NA 7 NA NA 6 5 ...
##   ..- attr(*, "label")= chr "Length of time in US"
## $ DMDEDUC3: num NA NA NA 15 6 NA NA NA NA 12 ...
##   ..- attr(*, "label")= chr "Education level - Children/Youth 6-19"
## $ DMDEDUC2: num NA NA 2 NA NA 1 4 NA 5 NA ...
##   ..- attr(*, "label")= chr "Education level - Adults 20+"
## $ DMDMARTL: num NA NA 3 NA NA 1 2 NA 1 NA ...
##   ..- attr(*, "label")= chr "Marital status"
## $ RIDEXPRG: num NA NA NA NA NA NA NA NA NA NA ...
##   ..- attr(*, "label")= chr "Pregnancy status at exam"
## $ SIALANG : num 1 1 1 1 1 1 1 1 1 1 ...
##   ..- attr(*, "label")= chr "Language of SP Interview"
## $ SIAPROXY: num 1 1 2 2 1 2 2 1 2 2 ...
##   ..- attr(*, "label")= chr "Proxy used in SP Interview?"
## $ SIAINTRP: num 2 2 2 2 2 1 2 2 2 2 ...
##   ..- attr(*, "label")= chr "Interpreter used in SP Interview?"
## $ FIALANG : num 1 1 1 NA 1 1 1 1 1 2 ...
##   ..- attr(*, "label")= chr "Language of Family Interview"
## $ FIAPROXY: num 2 2 2 NA 2 2 2 2 2 2 ...
##   ..- attr(*, "label")= chr "Proxy used in Family Interview?"
## $ FIAINTRP: num 2 2 2 NA 2 2 2 2 2 2 ...
##   ..- attr(*, "label")= chr "Interpreter used in Family Interview?"
## $ MIALANG : num NA NA 1 1 1 1 NA NA 1 1 ...
##   ..- attr(*, "label")= chr "Language of MEC Interview"
## $ MIAPROXY: num NA NA 2 2 2 2 NA NA 2 2 ...
##   ..- attr(*, "label")= chr "Proxy used in MEC Interview?"
## $ MIAINTRP: num NA NA 2 2 2 1 NA NA 2 2 ...
##   ..- attr(*, "label")= chr "Interpreter used in MEC Interview?"
## $ AIALANGA: num NA NA 1 1 1 3 NA NA 1 1 ...
##   ..- attr(*, "label")= chr "Language of ACASI Interview"
## $ DMDHHSIZ: num 5 4 1 5 7 2 1 3 3 4 ...
##   ..- attr(*, "label")= chr "Total number of people in the Household"

```

```
## $ DMDFMSIZ: num 5 4 1 5 7 2 1 3 3 4 ...
##   ..- attr(*, "label")= chr "Total number of people in the Family"
## $ DMDHHSZA: num 3 2 0 0 0 0 1 0 0 ...
##   ..- attr(*, "label")= chr "# of children 5 years or younger in HH"
## $ DMDHHSZB: num 0 0 0 0 3 0 0 0 0 2 ...
##   ..- attr(*, "label")= chr "# of children 6-17 years old in HH"
## $ DMDHHSZE: num 0 0 1 1 0 2 1 0 0 0 ...
##   ..- attr(*, "label")= chr "# of adults 60 years or older in HH"
## $ DMDHRGND: num 1 1 2 1 1 1 2 1 1 2 ...
##   ..- attr(*, "label")= chr "HH ref person's gender"
## $ DMDHRAGZ: num 2 2 4 4 3 4 4 2 3 3 ...
##   ..- attr(*, "label")= chr "HH ref person's age in years"
## $ DMDHREDZ: num 3 3 1 3 2 1 2 3 3 1 ...
##   ..- attr(*, "label")= chr "HH ref person's education level"
## $ DMDHRMAZ: num 1 1 2 1 1 1 2 1 1 2 ...
##   ..- attr(*, "label")= chr "HH ref person's marital status"
## $ DMDHSEDZ: num 3 2 NA 2 3 1 NA 3 3 NA ...
##   ..- attr(*, "label")= chr "HH ref person's spouse's education level"
## $ WTINT2YR: num 9246 37339 8615 8549 6769 ...
##   ..- attr(*, "label")= chr "Full sample 2 year interview weight"
## $ WTMEC2YR: num 8540 42567 8338 8723 7065 ...
##   ..- attr(*, "label")= chr "Full sample 2 year MEC exam weight"
## $ SDMVPSU : num 2 1 2 2 1 2 1 1 2 2 ...
##   ..- attr(*, "label")= chr "Masked variance pseudo-PSU"
## $ SDMVSTRA: num 145 143 145 134 138 138 136 134 134 147 ...
##   ..- attr(*, "label")= chr "Masked variance pseudo-stratum"
## $ INDHHIN2: num 15 15 3 NA 10 6 2 15 15 4 ...
##   ..- attr(*, "label")= chr "Annual household income"
## $ INDFMIN2: num 15 15 3 NA 10 6 2 15 15 4 ...
##   ..- attr(*, "label")= chr "Annual family income"
## $ INDFMPIR: num 5 5 0.82 NA 1.88 1.63 0.41 4.9 5 0.76 ...
##   ..- attr(*, "label")= chr "Ratio of family income to poverty"
```

Variables Requiring Different Formats

The datasets contain a combination of numeric (num) or integer (int) values which are appropriate for continuous measurements like blood pressure and age. However, some variables are categorical variables that happen to be represented by numbers like the multiple language variables, as well as gender or marital status and will need to be converted during cleaning and recoding. However, since exercise 4 specifically asks to recode Gender and Marital status, I'll use those as my examples of variables that are not in the correct format.

Variable	Current type	Recommended type	Reason
RIAGENDR	Numeric	Factor	Represents gender groups (categories) rather than quantities or numerical measurement. Treating these codes like actual numbers could lead to inappropriate calculations or interpretations.

Variable	Current type	Recommended type	Reason
DMDMARTL	Numeric	Factor	Represents marital-status categories rather than quantities or numerical measurement. Treating these codes like actual numbers could lead to inappropriate calculations or interpretations.

Exercise 3 Merging and preparing datasets

What are the dimension of the final merged dataset?

29,400 observations and 19 variables ($29,400 \times 19$).

How do these dimension compare to the dimensions of the combined datasets used to create it?

The combined blood pressure dataset contains 28,061 observations and 11 variables, while the combined demographic dataset contains 29,400 observations and 10 variables. After merging, the final dataset contains 29,400 observations and 19 variables.

The number of observations remains 29,400 because the demographic dataset contains 29,400 participants, while the blood pressure data represent a subset of these participants. The `full_join()` retains all observations from both datasets, including participants who do not have blood pressure measurements.

The number of variables increases from 10 and 11 to 19 because the two datasets are combined using `SEQN` and `wave_id` as matching variables. These two variables are shared by both datasets and only included once in the final dataset. Thus, the number of variables is calculated as: $10 + 11 - 2 = 19$.

Exercise 4 Data cleaning and variable recoding

```
# Skim data
skim(nhanes_merged)
```

Table 1: Data summary

Name	nhanes_merged
Number of rows	29400
Number of columns	19
Column type frequency:	
character	1
numeric	18

Group variables	None
-----------------	------

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
wave_id	0	1	9	9	0	3	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
SEQN	0	1.00	88256.50	8487.19	73557	80906.75	88256.5	95606.25	102956	
RIAGENDR	0	1.00	1.51	0.50	1	1.00	2.0	2.00	2	
RIDAGEYR	0	1.00	32.52	24.91	0	10.00	28.0	54.00	80	
RIDRETH1	0	1.00	3.11	1.29	1	2.00	3.0	4.00	5	
RIDRETH3	0	1.00	3.33	1.67	1	2.00	3.0	4.00	7	
DMDEDUC2	12343	0.58	3.49	1.26	1	3.00	4.0	4.00	9	
DMDMARTL	12343	0.58	2.64	2.79	1	1.00	1.0	5.00	99	
DMDFMSIZ	0	1.00	3.70	1.78	1	2.00	4.0	5.00	7	
INDFMIN2	926	0.97	11.28	15.34	1	5.00	8.0	14.00	99	
PEASCCT1	28400	0.03	20.64	30.89	1	3.00	3.0	56.00	122	
BPXSY1	8781	0.70	119.94	18.91	66	106.00	116.0	130.00	236	
BPXDI1	8781	0.70	66.55	15.20	0	58.00	68.0	76.00	136	
BPXSY2	8115	0.72	119.99	19.11	66	106.00	116.0	130.00	238	
BPXDI2	8115	0.72	66.46	15.46	0	58.00	68.0	76.00	144	
BPXSY3	8178	0.72	119.67	18.90	62	106.00	116.0	130.00	238	
BPXDI3	8178	0.72	66.36	15.75	0	58.00	68.0	76.00	140	
BPXSY4	28005	0.05	128.18	23.92	74	110.00	126.0	144.00	234	
BPXDI4	28005	0.05	71.30	17.27	0	62.00	72.0	82.00	132	

```
## Clean column names
```

```
nhanes_clean <- nhanes_merged %>%
  clean_names()
```

```
names(nhanes_clean)
```

```
## [1] "seqn"      "riagendr"  "ridageyr"  "ridreth1"  "ridreth3"  "dmdeduc2"
## [7] "dmdmartl"  "dmdfmsiz"  "indfmin2"  "wave_id"   "peascct1"  "bpxsy1"
## [13] "bpxdi1"    "bpxsy2"    "bpxdi2"    "bpxsy3"    "bpxdi3"    "bpxsy4"
## [19] "bpxdi4"
```

```
nhanes_clean <- nhanes_clean %>%
  rename(
    participant_id = seqn,
    gender = riagendr,
    age_years = ridageyr,
    ethnicity_original = ridreth1,
    ethnicity_asian = ridreth3,
    education = dmdeduc2,
```

```

marital_status = dmdmart1,
family_size = dmdfmsiz,
family_income = indfmin2,
bp_comment = peascct1,
systolic_1 = bpxsy1,
diastolic_1 = bpxdi1,
systolic_2 = bpxsy2,
diastolic_2 = bpxdi2,
systolic_3 = bpxsy3,
diastolic_3 = bpxdi3,
systolic_4 = bpxsy4,
diastolic_4 = bpxdi4
)

## Standardize and recode variables
nhanes_clean <- nhanes_clean %>%
  mutate(
    gender = case_when(
      gender == 1 ~ "Male",
      gender == 2 ~ "Female",
      TRUE ~ NA_character_
    ),
    gender = factor(gender)
  )

table(nhanes_clean$gender, useNA = "ifany")

```

```

##
## Female    Male
## 14948    14452

```

```

nhanes_clean <- nhanes_clean %>%
  mutate(
    ethnicity = case_when(
      ethnicity_asian == 1 ~ "Mexican American",
      ethnicity_asian == 2 ~ "Other Hispanic",
      ethnicity_asian == 3 ~ "Non-Hispanic White",
      ethnicity_asian == 4 ~ "Non-Hispanic Black",
      ethnicity_asian == 6 ~ "Non-Hispanic Asian",
      ethnicity_asian == 7 ~ "Other Race/Multi-Racial",
      TRUE ~ NA_character_
    ),
    ethnicity = factor(ethnicity)
  )

table(nhanes_clean$ethnicity, useNA = "ifany")

```

```

##
##      Mexican American      Non-Hispanic Asian      Non-Hispanic Black
##              5018              3284              6511
##      Non-Hispanic White      Other Hispanic Other Race/Multi-Racial
##              9890              3088              1609

```

```

nhanes_clean <- nhanes_clean %>%
  mutate(
    education = case_when(
      education == 1 ~ "Less than 9th grade",
      education == 2 ~ "9th-11th grade",
      education == 3 ~ "High school graduate/GED",
      education == 4 ~ "Some college/AA degree",
      education == 5 ~ "College graduate or higher",
      education %in% c(7, 9) ~ NA_character_,
      TRUE ~ NA_character_
    ),
    education = factor(
      education,
      levels = c(
        "Less than 9th grade",
        "9th-11th grade",
        "High school graduate/GED",
        "Some college/AA degree",
        "College graduate or higher"
      ),
      ordered = TRUE
    )
  )

table(nhanes_clean$education, useNA = "ifany")

```

```

##
##      Less than 9th grade      9th-11th grade
##              1622              2105
##  High school graduate/GED  Some college/AA degree
##              3864              5240
## College graduate or higher      <NA>
##              4201              12368

```

```

nhanes_clean <- nhanes_clean %>%
  mutate(
    marital_status = case_when(
      marital_status == 1 ~ "Married",
      marital_status == 2 ~ "Widowed",
      marital_status == 3 ~ "Divorced",
      marital_status == 4 ~ "Separated",
      marital_status == 5 ~ "Never married",
      marital_status == 6 ~ "Living with partner",
      marital_status %in% c(77, 99) ~ NA_character_,
      TRUE ~ NA_character_
    ),
    marital_status = factor(marital_status)
  )

table(nhanes_clean$marital_status, useNA = "ifany")

```

```

##

```

##	Divorced Living with partner	Married	Never married
##	1914	1487	8588
##	Separated	Widowed	<NA>
##	571	1319	12355

```
nhanes_clean <- nhanes_clean %>%
  mutate(
    family_income = case_when(
      family_income %in% c(1, 2, 3, 4) ~ "Under $20,000",
      family_income %in% c(5, 6, 7) ~ "$20,000-$44,999",
      family_income %in% c(8, 9, 10) ~ "$45,000-$74,999",
      family_income == 14 ~ "$75,000-$99,999",
      family_income == 15 ~ "$100,000 or more",
      family_income %in% c(12, 13) ~ NA_character_,
      family_income %in% c(77, 99) ~ NA_character_,
      TRUE ~ NA_character_
    ),
    family_income = factor(
      family_income,
      levels = c(
        "Under $20,000",
        "$20,000-$44,999",
        "$45,000-$74,999",
        "$75,000-$99,999",
        "$100,000 or more"
      ),
      ordered = TRUE
    )
  )

table(nhanes_clean$family_income, useNA = "ifany")
```

##	Under \$20,000	\$20,000-\$44,999	\$45,000-\$74,999	\$75,000-\$99,999
##	5843	8066	5074	2524
##	\$100,000 or more	<NA>		
##	4825	3068		

```
nhanes_clean <- nhanes_clean %>%
  mutate(
    family_size_category = case_when(
      family_size <= 2 ~ "1-2 people",
      family_size <= 4 ~ "3-4 people",
      family_size >= 5 ~ "5+ people",
      TRUE ~ NA_character_
    ),
    family_size_category = factor(
      family_size_category,
      levels = c(
        "1-2 people",
        "3-4 people",
        "5+ people"
      ),

```

```

    ordered = TRUE
  )
)

table(nhanes_clean$family_size_category, useNA = "ifany")

```

```

##
## 1-2 people 3-4 people 5+ people
##      8689      10826      9885

```

```

nhanes_clean <- nhanes_clean %>%
  mutate(
    age_category = case_when(
      age_years <= 17 ~ "0-17",
      age_years <= 29 ~ "18-29",
      age_years <= 44 ~ "30-44",
      age_years <= 59 ~ "45-59",
      age_years <= 74 ~ "60-74",
      age_years >= 75 ~ "75+",
      TRUE ~ NA_character_
    ),
    age_category = factor(
      age_category,
      levels = c(
        "0-17",
        "18-29",
        "30-44",
        "45-59",
        "60-74",
        "75+"
      )
    ),
    ordered = TRUE
  )

table(nhanes_clean$age_category, useNA = "ifany")

```

```

##
## 0-17 18-29 30-44 45-59 60-74 75+
## 11439 3662 4267 4140 4033 1859

```

```

## Remove duplicate entries
sum(duplicated(nhanes_clean))

```

```
## [1] 0
```

```

nhanes_clean <- nhanes_clean %>%
  distinct()

sum(duplicated(nhanes_clean))

```

```
## [1] 0
```

```
## Final data checks
dim(nhanes_clean)
```

```
## [1] 29400    22
```

```
names(nhanes_clean)
```

```
## [1] "participant_id"      "gender"      "age_years"
## [4] "ethnicity_original"  "ethnicity_asian" "education"
## [7] "marital_status"     "family_size"  "family_income"
## [10] "wave_id"            "bp_comment"   "systolic_1"
## [13] "diastolic_1"        "systolic_2"   "diastolic_2"
## [16] "systolic_3"         "diastolic_3"  "systolic_4"
## [19] "diastolic_4"        "ethnicity"    "family_size_category"
## [22] "age_category"
```

```
skim(nhanes_clean)
```

Table 2: Data summary

Name	nhanes_clean
Number of rows	29400
Number of columns	22
Column type frequency:	
character	1
factor	7
numeric	14
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
wave_id	0	1	9	9	0	3	0

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
gender	0	1.00	FALSE	2	Fem: 14948, Mal: 14452
education	12368	0.58	TRUE	5	Som: 5240, Col: 4201, Hig: 3864, 9th: 2105
marital_status	12355	0.58	FALSE	6	Mar: 8588, Nev: 3166, Div: 1914, Liv: 1487
family_income	3068	0.90	TRUE	5	\$20: 8066, Und: 5843, \$45: 5074, \$10: 4825
ethnicity	0	1.00	FALSE	6	Non: 9890, Non: 6511, Mex: 5018, Non: 3284

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
family_size_category	0	1.00	TRUE	3	3-4: 10826, 5+ : 9885, 1-2: 8689
age_category	0	1.00	TRUE	6	0-1: 11439, 30-: 4267, 45-: 4140, 60-: 4033

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
participant_id	0	1.00	88256.50	8487.19	73557	80906.75	88256.5	95606.25	102956	
age_years	0	1.00	32.52	24.91	0	10.00	28.0	54.00	80	
ethnicity_original	0	1.00	3.11	1.29	1	2.00	3.0	4.00	5	
ethnicity_asian	0	1.00	3.33	1.67	1	2.00	3.0	4.00	7	
family_size	0	1.00	3.70	1.78	1	2.00	4.0	5.00	7	
bp_comment	28400	0.03	20.64	30.89	1	3.00	3.0	56.00	122	
systolic_1	8784	0.70	119.94	18.91	66	106.00	116.0	130.00	236	
diastolic_1	8784	0.70	66.55	15.20	0	58.00	68.0	76.00	136	
systolic_2	8120	0.72	119.99	19.11	66	106.00	116.0	130.00	238	
diastolic_2	8120	0.72	66.46	15.46	0	58.00	68.0	76.00	144	
systolic_3	8182	0.72	119.66	18.87	62	106.00	116.0	130.00	232	
diastolic_3	8182	0.72	66.37	15.74	0	58.00	68.0	76.00	140	
systolic_4	28007	0.05	128.16	23.81	76	110.00	126.0	144.00	234	
diastolic_4	28007	0.05	71.30	17.27	0	62.00	72.0	82.00	132	

```
## Export cleaned dataset
export(
  nhanes_clean,
  here("NHANES Data", "NHANES_BP_2013_2018_cleaned.csv")
)
```